

Draft Provisional Patent Application (Outline)

Title:

System and Method for Embedding Voice-to-Voice Artificial Intelligence into Website Navigation with Abuse Detection, SKU Integration, and Memory-Linked Personalization

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1. Field of the Invention

The present invention relates to conversational artificial intelligence, specifically a system and method for embedding real-time, voice-to-voice AI functionality directly into website navigation toolbars for commerce, customer service, and information retrieval.

2. Background

Current AI chatbots primarily rely on text-based “chat bubbles” or standalone apps. These approaches lack brand integration, create trust barriers, and often fail under noisy environments or with multiple concurrent users. Existing solutions do not effectively combine:

- Toolbar-level embedding,
- Abuse detection with enforced manual fallback,
- SKU-based transaction flows,

- Personalized memory tied to app or user identifiers,
 - And transcript-based SEO/data benefits.
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3. Summary of the Invention

This invention provides a **system and method** that:

1. Embeds an AI voice assistant directly into a website toolbar for seamless brand trust.
 2. Processes user speech through an English-only ASR pipeline with noise suppression.
 3. Detects abusive or nonsensical use and automatically reverts to manual human-controlled mode, optionally enforcing temporary lockouts.
 4. Maps transcripts to SKU libraries, enabling voice-driven commerce transactions (search, add to cart, checkout).
 5. Stores and recalls memory tied to user IDs or app logins, enabling personalized repeat interactions.
 6. Exports transcripts for SEO indexing and training data improvement.
 7. Supports “Mothership + Satellite” architecture where core AI improvements flow downward, but client proprietary data remains firewalled.
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4. Detailed Description (Plain Language)

- **Toolbar Integration:** A code snippet embeds a microphone button into the site's navigation bar, ensuring the assistant inherits the credibility of the host brand rather than appearing as a third-party widget.
- **Noise Suppression Pipeline:** Audio is streamed to an English-only automatic speech recognition (ASR) service (e.g., Deepgram) that blocks unintended languages and filters

background noise (music, chatter, engines).

- **Abuse Detection:** If transcripts reveal trolling, nonsense, or malicious intent, the system locks the user into manual mode for a set period, preventing resource waste or misorders.
 - **SKU Mapping:** The assistant converts natural language (“olive oil, 16 oz, cheapest”) into SKU-based selections, adding items to cart or initiating checkout.
 - **Memory Linkage:** Repeat users are recognized via app or site identifiers, allowing the AI to recall past orders or preferences.
 - **Transcript Export:** Conversations are logged and available for search engine indexing, enhancing site SEO and AI search visibility.
 - **Mothership–Satellite Architecture:** A core AI model (“Mothership”) pushes generic improvements (latency, voice quality, reasoning) to all satellites (e.g., Walmart instance, CorvetteParts instance). Proprietary client data (menus, SKU lists) is isolated by firewall, never returning to the mothership.
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5. Claims (Plain Draft)

1. A system for embedding a real-time voice assistant into a website toolbar comprising:
 - a microphone interface,
 - a streaming ASR pipeline constrained to English with noise suppression,
 - and a WebRTC connection to a reasoning model providing audio responses.
2. The system of claim 1 further comprising abuse detection logic that triggers enforced manual fallback mode and temporary lockout of abusive users.
3. The system of claim 1 further comprising SKU mapping from transcripts to product databases for e-commerce transactions.
4. The system of claim 1 further comprising session memory tied to application or user identifiers for personalized repeat interactions.

5. The system of claim 1 further comprising transcript export to external search engines for SEO enhancement.
 6. The system of claim 1 implemented in a mothership–satellite architecture where generic improvements are shared downstream and client proprietary data is firewalled upstream.
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6. Abstract

A system and method for embedding real-time, voice-to-voice artificial intelligence directly into website navigation toolbars. The system integrates speech recognition, abuse detection, SKU mapping, personalized memory, and transcript export into a commerce-ready AI. Proprietary client data remains isolated within a mothership–satellite architecture.